Bihar Engineering University, Patna

B.Tech 7th Semester Examination, 2024

Course: B.Tech
Code: 104710

Subject: VLSI Technology

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin
*(ii) There are **NINE** questions in this paper*
*(iii) Attempt **FIVE** questions in all*
(iv) Question No. 1 is compulsory

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which of the following is an advantage of CMOS technology over NMOS technology?
(i) Higher power dissipation
(ii) Lower power consumption
(iii) Lower noise margin
(iv) Requires only NMOS transistors

(b) Which stage of VLSI design is responsible for defining the overall system structure and functionality?
(i) Logic Design
(ii) Architecture Definition
(iii) Circuit Design
(iv) Physical Design

(c) Which crystal growth technique is most commonly used for producing high-purity silicon wafers in the semiconductor industry?
(i) Bridgman method
(ii) Hydrothermal growth
(iii) Czochralski technique
(iv) Vapor-phase epitaxy

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(d) What drives the diffusion process in a semiconductor material?
(i) Electric field
(ii) Concentration gradient
(iii) Magnetic field
(iv) Mechanical stress

(e) What is a major issue in epitaxial growth that can lead to defects in the crystal structure?
(i) Low doping concentration
(ii) Excessive impurity diffusion
(iii) Dislocations due to lattice mismatch
(iv) Rapid cooling of the wafer

(f) Which factor primarily determines the penetration range of ions during ion implantation in semiconductors?
(i) Ion mass
(ii) Implantation angle
(iii) Ion energy
(iv) Crystal orientation

(g) What is the primary factor that influences the rate of thermal oxidation of silicon?
(i) Temperature
(ii) Thickness of the silicon wafer
(iii) Oxygen pressure
(iv) Both (i) and (iii)

(h) Which film deposition method involves the physical transfer of material from a solid target to the substrate in a vacuum environment?
(i) Chemical Vapor Deposition
(ii) Sputter Deposition
(iii) Vacuum Evaporation
(iv) Atomic Layer Deposition

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(i) Which etching technique is most commonly used for high precision patterning in modern semiconductor fabrication?

(i) Wet chemical etching
(ii) Dry chemical etching
(iii) Dry physical etching
(iv) Reactive Ion Etching

(j) What is the main purpose of planarization in semiconductor processing?

(i) To increase the doping concentration
(ii) To create a smooth and uniform surface for subsequent processing steps
(iii) To reduce the number of defects during oxide growth
(iv) To prevent metal contamination

Q.2 Explain the fundamental differences between the NMOS and CMOS fabrication processes. Compare the P-Well, N-Well, and Twin-Tub processes in CMOS technology. [14]

Q.3 Explain the different stages of VLSI design, starting from Architecture Definition to Physical Design. How do these stages contribute to the final implementation of an integrated circuit? [14]

Q.4 Explain the different doping methods used in semiconductor crystal growth, including rapid stirring conditions, partial stirring conditions and neutron transmutation doping. How do these methods influence the electrical properties of the crystal? [14]

Q.5 Discuss the impact of oxide growth on impurity redistribution in silicon. How does diffusion during oxide growth affect device performance and what are the key techniques used to evaluate diffused layers? [14]

Q.6 Discuss the effects of thermally induced strain and dislocations in epitaxial growth. How do these factors impact the quality and performance of semiconductor devices? [14]

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Q.7 Explain the ion implantation process, including the concepts of nuclear and electronic stopping. How do these stopping mechanisms affect the penetration range of implanted ions? [14]

Q.8 Describe the role of lithography in semiconductor fabrication. Discuss the process of pattern generation, mask making & pattern transfer and explain how advanced techniques like short-wavelength lithography, multilayer resist, phase-shifting masks & electron beam techniques improve pattern resolution and transfer accuracy. [14]

Q.9 Discuss the process flow for CMOS fabrication, highlighting key steps like self-alignment, metallization, gettering and planarization. How does each step influence the performance, reliability and yield of the final CMOS device? [14]

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